

Q. Discuss the Interrupt structure of the 8086 microprocessor. Differentiate between Hardware and Software interrupts, and explain the sequence of steps the CPU follows to service an interrupt.

📌 Definition to Remember

An **Interrupt** pauses the CPU's normal execution to handle an urgent task (an Interrupt Service Routine or ISR). The 8086 supports 256 interrupts. It uses an **Interrupt Vector Table (IVT)** in lower memory to locate the ISR addresses, ensuring the CPU state is saved on the stack before jumping.

1. Hardware vs. Software Interrupts

Type	Source	Characteristics	Examples
Hardware	External physical devices	Triggered by electrical signals to CPU pins. Can be Maskable (ignored if IF=0) or Non-Maskable.	INTR (Maskable), NMI (Non-Maskable).
Software	Internal program execution	Triggered by the <code>INT n</code> instruction or internal CPU exceptions. Cannot be masked.	<code>INT 21H</code> (DOS), Divide-by-zero (Type 0).

2. The Interrupt Vector Table (IVT)

- Located at absolute memory `00000H` to `003FFH`.
- Supports 256 interrupt types (Type 0 to 255).
- Each type gets exactly **4 bytes** in the IVT (2 bytes for the ISR's `IP` offset, 2 bytes for its `CS` segment).

3. Interrupt Servicing Sequence

When a valid interrupt is acknowledged, the 8086 performs the following strict sequence to ensure it doesn't lose its place in the main program:

1. **Push Flags:** Pushes the 16-bit Flag Register onto the Stack.
2. **Clear Flags:** Clears the Interrupt Flag (IF=0) and Trap Flag (TF=0) to prevent other maskable interrupts from interrupting the ISR.
3. **Save Context:** Pushes the current Code Segment (CS) and Instruction Pointer (IP) onto the Stack (saves the return address).
4. **Fetch ISR Address:** Multiplies the interrupt type number by 4 to find its IVT entry. Loads the new CS and IP values.
5. **Execute ISR:** Jumps to and executes the Interrupt Service Routine.
6. **Return (IRET):** The ISR ends with the `IRET` instruction, which pops the IP, CS, and Flags back from the stack, perfectly resuming the main program.

★ Must-Write Points (for 10 marks)

1. An **Interrupt** forces the CPU to pause execution and run a specific routine (ISR).
2. **Hardware Interrupts:** External signals. INTR is maskable; NMI is non-maskable (for critical errors).
3. **Software Interrupts:** Triggered by `INT n` instructions or exceptions (divide-by-zero).
4. **Interrupt Vector Table (IVT):** Occupies the lowest 1KB of memory (00000H-003FFH).

5. IVT holds 256 entries. Each entry is 4 bytes (CS and IP for the ISR).

6. **Servicing Sequence:** Push Flags → Clear IF/TF → Push CS and IP → Load new CS/IP from IVT → Execute ISR.

7. The ISR must end with **IRET** to pop the IP, CS, and Flags from the stack and resume the program.

⚡ Quick Recall

Interrupts → Hardware (INTR/NMI) vs Software (INT n) → IVT (Lowest memory, 256 types, 4 bytes each for CS:IP) → Sequence: Push Flags → Clear IF/TF → Push CS:IP → Fetch ISR from IVT → Run ISR → IRET (Pop all back)